

```

# Enhanced Calendar System Configuration
# File: enhanced_config.yaml

# =====
# CORE SYSTEM CONFIGURATION
# =====

# Calendar Import Settings
calendar_auto_import: true
import_day: 6 # Sunday (0=Monday, 6=Sunday)
import_hour: 12 # 12 PM CST
retry_interval_hours: 1
max_retry_attempts: 24
import_timeout_seconds: 30

# File Processing Settings
chunk_size: 1000
memory_limit_mb: 100
max_workers: 10
max_connections: 10

# Event Processing Settings
anticipation_hours: [1, 2, 4]
anticipation_enabled: true
minimum_gap_minutes: 30

# Trigger Configuration
trigger_offsets:
    EMO-E: -3 # High impact: -3 minutes
    EMO-A: -2 # Medium impact: -2 minutes
    EQT-OPEN: -5 # Equity markets: -5 minutes
    ANTICIPATION: -1 # Anticipation events: -1 minute

# Parameter Sets (4 rotation sets)
parameter_sets:
    1:
        lot_size: 0.01
        buy_distance: 20
        sell_distance: 20
        stop_loss: 40
        take_profit: 80
        risk_level: "low"
    2:
        lot_size: 0.02

```

```

    buy_distance: 20
    sell_distance: 20
    stop_loss: 40
    take_profit: 80
    risk_level: "medium"
3:
    lot_size: 0.03
    buy_distance: 20
    sell_distance: 20
    stop_loss: 40
    take_profit: 80
    risk_level: "medium"
4:
    lot_size: 0.04
    buy_distance: 20
    sell_distance: 20
    stop_loss: 40
    take_profit: 80
    risk_level: "high"

# =====
# DATABASE CONFIGURATION
# =====

database_path: "data/enhanced_calendar.db"
backup_retention_days: 30
auto_vacuum: true
wal_mode: true

# =====
# NOTIFICATION SYSTEM CONFIGURATION
# =====

notification_settings:
    email_enabled: true
    sms_enabled: true
    webhook_enabled: true

# SMTP Configuration for Email
smtp_settings:
    host: "smtp.gmail.com"
    port: 587
    username: "your-email@gmail.com"
    password: "your-app-password" # Use app-specific password
    use_tls: true

# Twilio Configuration for SMS

```

```

twilio_settings:
    account_sid: "your-twilio-account-sid"
    auth_token: "your-twilio-auth-token"
    from_number: "+1234567890"

# Webhook Configuration
webhook_settings:
    endpoints:
        - url: "https://your-webhook-endpoint.com/alerts"
          secret: "your-webhook-secret"

# Alert Recipients
alert_recipients:
    - "admin@yourdomain.com"
    - "+1234567890" # SMS number

# Notification Priorities
critical_events:
    - "system_startup"
    - "system_shutdown"
    - "database_error"
    - "memory_overflow"
    - "circuit_breaker_open"

high_priority_events:
    - "signal_generation_error"
    - "file_processing_error"
    - "network_timeout"

medium_priority_events:
    - "configuration_change"
    - "backtest_completion"
    - "compliance_violation"

# =====
# RISK MANAGEMENT CONFIGURATION
# =====

risk_management:
    # Global Risk Limits
    max_daily_risk_pct: 5.0 # Maximum 5% of account per day
    max_portfolio_risk_pct: 10.0 # Maximum 10% of account total
    max_trades_per_day: 10
    max_trades_per_hour: 3

    # Position Sizing
    default_risk_per_trade_pct: 2.0 # 2% risk per trade

```

```
min_lot_size: 0.01
max_lot_size: 1.0

# Correlation Limits
max_correlation_exposure: 0.7 # Maximum 70% correlation between positions
correlation_window_days: 30

# Drawdown Protection
max_drawdown_pct: 15.0 # Stop trading at 15% drawdown
daily_loss_limit_pct: 3.0 # Stop trading at 3% daily loss

# Currency Exposure Limits
currency_limits:
    USD: 50.0 # Maximum 50% exposure to USD pairs
    EUR: 30.0
    GBP: 20.0
    JPY: 20.0
    AUD: 15.0
    CAD: 15.0
    CHF: 10.0
    NZD: 10.0

# =====
# PORTFOLIO MANAGEMENT CONFIGURATION
# =====

portfolio_management:
    # Account Configuration
    accounts:
        account_1:
            account_id: "live_account_1"
            account_type: "live"
            currency: "USD"
            initial_balance: 10000
            risk_profile: "conservative"
            active: true
            max_lot_size: 0.5
            allowed_symbols: ["EURUSD", "GBPUSD", "USDJPY", "USDCAD"]

        account_2:
            account_id: "demo_account_1"
            account_type: "demo"
            currency: "EUR"
            initial_balance: 5000
            risk_profile: "aggressive"
            active: true
            max_lot_size: 1.0
```

```

    allowed_symbols: ["EURUSD", "GBPUSD", "EURGBP", "EURJPY"]

# Risk Profile Settings
risk_profiles:
    conservative:
        risk_multiplier: 0.5
        max_daily_trades: 5
        allowed_risk_levels: ["low"]

    moderate:
        risk_multiplier: 1.0
        max_daily_trades: 8
        allowed_risk_levels: ["low", "medium"]

    aggressive:
        risk_multiplier: 1.5
        max_daily_trades: 12
        allowed_risk_levels: ["low", "medium", "high"]

# =====
# BACKTESTING CONFIGURATION
# =====

backtesting:
    # Default Backtest Settings
    initial_balance: 10000
    commission_per_lot: 7.0 # $7 per lot round trip
    spread_pips: 1.5 # Average spread in pips
    slippage_pips: 0.5 # Average slippage in pips

    # Data Sources
    historical_data_path: "data/historical/"
    supported_symbols: ["EURUSD", "GBPUSD", "USDJPY", "USDCAD", "AUDUSD", "NZDUSD",

    # Analysis Settings
    calculate_sharpe: true
    calculate_sortino: true
    calculate_calmar: true
    benchmark_symbol: "EURUSD"

    # Report Generation
    generate_charts: true
    chart_output_path: "reports/charts/"
    pdf_reports: true
    pdf_output_path: "reports/pdf/"

# =====

```

```

# COMPLIANCE CONFIGURATION
# =====

compliance:
    # Regulatory Settings
    jurisdiction: "US" # US, EU, UK, etc.
    mifid_ii_compliance: false
    dodd_frank_compliance: true

    # Audit Trail Settings
    log_all_trades: true
    log_all_signals: true
    log_configuration_changes: true
    audit_retention_days: 2555 # 7 years

    # Reporting Settings
    daily_reports: true
    weekly_reports: true
    monthly_reports: true
    quarterly_reports: true

    # Report Recipients
    compliance_recipients:
        - "compliance@yourdomain.com"
        - "risk@yourdomain.com"

    # Violation Thresholds
    position_size_limit: 1.0 # Maximum 1.0 lots per position
    daily_loss_threshold: 500 # $500 daily loss threshold
    weekly_loss_threshold: 1500 # $1500 weekly loss threshold

    # Required Reports
    required_monthly_metrics:
        - "total_trades"
        - "win_rate"
        - "profit_factor"
        - "max_drawdown"
        - "sharpe_ratio"
        - "var_95" # Value at Risk 95%
        - "position_concentration"

# =====
# MONITORING & PERFORMANCE CONFIGURATION
# =====

monitoring:
    # Health Check Settings

```

```
health_check_interval: 30 # seconds
performance_monitoring: true
memory_monitoring: true
disk_monitoring: true

# Performance Thresholds
max_memory_usage_pct: 80
max_cpu_usage_pct: 85
max_disk_usage_pct: 90
max_database_response_ms: 1000

# Metrics Collection
collect_system_metrics: true
collect_trading_metrics: true
collect_error_metrics: true
metrics_retention_days: 90

# Alerting Thresholds
critical_thresholds:
    memory_usage_pct: 90
    cpu_usage_pct: 95
    disk_usage_pct: 95
    database_response_ms: 5000
    error_rate_per_hour: 10

warning_thresholds:
    memory_usage_pct: 80
    cpu_usage_pct: 85
    disk_usage_pct: 85
    database_response_ms: 1000
    error_rate_per_hour: 5

# =====
# INTEGRATION CONFIGURATION
# =====

integration:
    # MT4/MT5 Integration
    mt4_enabled: true
    mt4_signal_path: "signals/mt4_signals.csv"
    mt4_status_path: "status/mt4_status.csv"
    mt4_response_path: "responses/mt4_responses.csv"

    # Redis Integration (for distributed systems)
    redis_enabled: false
    redis_host: "localhost"
    redis_port: 6379
```

```
redis_db: 0
redis_password: ""

# MQTT Integration (for real-time messaging)
mqtt_enabled: false
mqtt_broker: "localhost"
mqtt_port: 1883
mqtt_topics:
    signals: "trading/signals"
    status: "trading/status"
    alerts: "trading/alerts"

# External APIs
external_apis:
    calendar_providers:
        forexfactory:
            enabled: true
            url: "https://www.forexfactory.com/calendar"
            backup_enabled: true

        investing_com:
            enabled: false
            api_key: ""

    market_data:
        alpha_vantage:
            enabled: false
            api_key: ""

        yahoo_finance:
            enabled: true

# =====
# SECURITY CONFIGURATION
# =====

security:
    # Encryption Settings
    encrypt_sensitive_data: true
    encryption_key_path: "keys/encryption.key"

    # Access Control
    require_authentication: false
    session_timeout_minutes: 60

    # API Security
    api_rate_limiting: true
```



```
max_requests_per_minute: 100

# File Security
secure_file_permissions: true
backup_encryption: true

# Logging Security
log_sensitive_data: false
anonymize_personal_data: true

# =====
# PATHS CONFIGURATION
# =====

paths:
    # Data Directories
    data_dir: "data/"
    calendar_import_path: "downloads/"
    signals_export_path: "signals/"
    status_export_path: "status/"
    backup_path: "backups/"
    logs_path: "logs/"
    reports_path: "reports/"

    # File Patterns
    calendar_file_patterns:
        - "ff_calendar*.csv"
        - "*thisweek*.csv"
        - "calendar*.csv"
        - "*forex*.csv"

    # Archive Settings
    archive_processed_files: true
    archive_path: "archive/"
    archive_retention_days: 90

# =====
# DEVELOPMENT & TESTING CONFIGURATION
# =====

development:
    # Debug Settings
    debug_mode: false
    verbose_logging: false
    log_level: "INFO" # DEBUG, INFO, WARNING, ERROR, CRITICAL

    # Testing Settings
```

```
test_mode: false
mock_trading: false
simulation_speed: 1.0 # 1.0 = real-time, 10.0 = 10x speed

# Development Tools
auto_reload: true
profiling_enabled: false
memory_profiling: false

# Test Data
test_data_path: "tests/data/"
generate_test_signals: false
test_notification_recipient: "test@yourdomain.com"

---

# Deployment Configuration
# File: docker-compose.yml

version: '3.8'

services:
  enhanced-calendar:
    build: .
    container_name: enhanced-calendar-system
    restart: unless-stopped
    volumes:
      - ./data:/app/data
      - ./logs:/app/logs
      - ./config:/app/config
      - ./downloads:/app/downloads
      - ./signals:/app/signals
      - ./backups:/app/backups
    environment:
      - PYTHONPATH=/app
      - LOG_LEVEL=INFO
    ports:
      - "8000:8000" # API port
    depends_on:
      - redis
      - mosquito
    healthcheck:
      test: ["CMD", "python", "-c", "import requests; requests.get('http://localh
      interval: 30s
      timeout: 10s
      retries: 3

  redis:
```

```
image: redis:7-alpine
container_name: calendar-redis
restart: unless-stopped
ports:
  - "6379:6379"
volumes:
  - redis_data:/data
command: redis-server --appendonly yes
```

```
mosquitto:
  image: eclipse-mosquitto:2
  container_name: calendar-mqtt
  restart: unless-stopped
  ports:
    - "1883:1883"
    - "9001:9001"
  volumes:
    - ./mosquitto.conf:/mosquitto/config/mosquitto.conf
    - mosquitto_data:/mosquitto/data
    - mosquitto_logs:/mosquitto/log
```

```
volumes:
  redis_data:
  mosquitto_data:
  mosquitto_logs:
```

```
# Requirements File
# File: requirements.txt
```

```
# Core Dependencies
asyncio-mqtt==0.13.0
aiofiles==23.2.1
aiohttp==3.9.1
aiosqlite==0.19.0
pydantic==2.5.0
structlog==23.2.0
watchdog==3.0.0
pyyaml==6.0.1
```

```
# Database & Performance
redis==5.0.1
lz4==4.3.2
psutil==5.9.6
```

```
# Notifications
twilio==8.10.0
```

```
jinja2==3.1.2

# Data Processing & Analysis
pandas==2.1.4
numpy==1.25.2
scipy==1.11.4

# Visualization & Reporting
plotly==5.17.0
matplotlib==3.8.2

# Backtesting & Statistics
ta-lib==0.4.28
statsmodels==0.14.0

# Web Framework (for API/dashboard)
fastapi==0.104.1
uvicorn==0.24.0

# Configuration & Validation
jsonschema==4.20.0
python-dotenv==1.0.0

# Testing
pytest==7.4.3
pytest-asyncio==0.21.1
pytest-cov==4.1.0

# Security & Encryption
cryptography==41.0.8
passlib==1.7.4

# Monitoring & Logging
prometheus-client==0.19.0
grafana-api==1.0.3

---
# Dockerfile
# File: Dockerfile

FROM python:3.11-slim

# Set working directory
WORKDIR /app

# Install system dependencies
RUN apt-get update && apt-get install -y \
```

```

gcc \
g++ \
make \
libc-dev \
libffi-dev \
&& rm -rf /var/lib/apt/lists/*

# Copy requirements first for better caching
COPY requirements.txt .

# Install Python dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY . .

# Create necessary directories
RUN mkdir -p data logs config downloads signals backups reports

# Set permissions
RUN chmod +x *.py

# Health check
HEALTHCHECK --interval=30s --timeout=10s --start-period=5s --retries=3 \
    CMD python -c "import requests; requests.get('http://localhost:8000/health')"

# Expose port
EXPOSE 8000

# Run the application
CMD ["python", "enhanced_calendar_system.py"]

---

# Installation & Deployment Guide
# File: DEPLOYMENT.md

# Enhanced Economic Calendar System - Deployment Guide

## Prerequisites

1. **Python 3.11+**
2. **Docker & Docker Compose** (for containerized deployment)
3. **Redis** (for distributed caching)
4. **MQTT Broker** (for real-time messaging)

## Quick Start (Docker)

```

1. ****Clone the repository:****
```bash  
git clone <repository-url>  
cd enhanced-calendar-system  
```
2. ****Configure the system:****
```bash  
cp enhanced\_config.yaml.example enhanced\_config.yaml  
# Edit enhanced\_config.yaml with your settings  
```
3. ****Start the system:****
```bash  
docker-compose up -d  
```
4. ****Verify deployment:****
```bash  
curl http://localhost:8000/health  
```

Manual Installation

1. ****Install Python dependencies:****
```bash  
pip install -r requirements.txt  
```
2. ****Initialize the database:****
```bash  
python -c "  
import asyncio  
from enhanced\_calendar\_system import OptimizedDatabaseManager  
async def init():  
 db = OptimizedDatabaseManager('data/enhanced\_calendar.db')  
 await db.initialize()  
asyncio.run(init())  
"  
```
3. ****Start the system:****
```bash  
python enhanced\_calendar\_system.py  
```

Configuration

Essential Configuration Items

1. **Database Path:**

```
```yaml
database_path: "data/enhanced_calendar.db"
```
```

2. **Notification Settings:**

```
```yaml
notification_settings:
 smtp_settings:
 host: "your-smtp-host"
 username: "your-email@domain.com"
 password: "your-password"
```
```

3. **Risk Management:**

```
```yaml
risk_management:
 max_daily_risk_pct: 5.0
 default_risk_per_trade_pct: 2.0
```
```

Environment Variables

Set these environment variables for enhanced security:

```
```bash
export SMTP_PASSWORD="your-smtp-password"
export TWILIO_AUTH_TOKEN="your-twilio-token"
export ENCRYPTION_KEY="your-encryption-key"
export DATABASE_PATH="/secure/path/to/database.db"
```
```

Performance Tuning

Database Optimization

1. **Increase cache size for better performance:**

```
```yaml
database_settings:
 cache_size: 10000
 mmap_size: 268435456 # 256MB
```
```

2. **Enable WAL mode for better concurrency:**

```

    ``yaml
database_settings:
    wal_mode: true
    synchronous: NORMAL
    ``

### Memory Management

1. **Adjust chunk size for large files:**
    ``yaml
    chunk_size: 1000 # Increase for more memory, decrease for less
    memory_limit_mb: 100 # Set based on available RAM
    ``

2. **Configure worker threads:**
    ``yaml
    max_workers: 10 # Set based on CPU cores
    max_connections: 10 # Database connection pool size
    ``

## Monitoring & Maintenance

### Health Monitoring

The system provides several health endpoints:

- `/health` - Basic health check
- `/metrics` - Prometheus metrics
- `/status` - Detailed system status

### Log Management

Logs are structured in JSON format and include:
- Component identification
- Error tracking with UUIDs
- Performance metrics
- Audit trails

### Backup Strategy

1. **Automated database backups:**
    ``yaml
    backup_retention_days: 30
    auto_backup_interval: "daily"
    ``

2. **Configuration backups:**

```


- Automatic backups before config changes
- Timestamped backup files
- Restore capabilities

Security Considerations

1. **File Permissions:**

```
```bash
chmod 600 enhanced_config.yaml
chmod 700 data/
chmod 700 logs/
```
```

2. **Network Security:**

- Use TLS for all external communications
- Implement API rate limiting
- Regular security updates

3. **Data Encryption:**

- Enable encryption for sensitive configuration data
- Encrypt database backups
- Secure key management

Troubleshooting

Common Issues

1. **Database Lock Errors:**

- Check for zombie processes
- Verify WAL mode is enabled
- Increase connection pool size

2. **Memory Issues:**

- Reduce chunk_size
- Lower memory_limit_mb
- Enable streaming processing

3. **Notification Failures:**

- Verify SMTP/SMS credentials
- Check network connectivity
- Review notification logs

Debug Mode

Enable debug mode for troubleshooting:

```
```yaml
```

```
development:
 debug_mode: true
 verbose_logging: true
 log_level: "DEBUG"
...
```

## ## Performance Benchmarks

### ### Expected Performance

- **File Processing:** 100+ events/second
- **Database Operations:** <10ms response time
- **Memory Usage:** <100MB for normal operation
- **Signal Generation:** <1 second latency

### ### Optimization Tips

1. Use SSD storage for database
2. Allocate sufficient RAM for caching
3. Monitor system resources regularly
4. Regular database maintenance

## ## Support & Maintenance

### ### Regular Maintenance Tasks

1. **Weekly:**
  - Review error logs
  - Check disk space
  - Verify backup integrity
2. **Monthly:**
  - Analyze performance metrics
  - Review compliance reports
  - Update dependencies
3. **Quarterly:**
  - Security audit
  - Performance tuning
  - Capacity planning

For additional support, check the documentation or contact the development team.